Author Disambiguation in Historical Academic Literature Using Large Language Models

Empirical Research Method

Term Paper

by

Sing-Hao Ku

Department of Economics

New York University

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Professor Sahar Parsa

Abstract

This paper addresses the critical problem of author name disambiguation in historical academic literature using Large Language Models (LLMs). Author name disambiguation—the task of correctly attributing publications to their true authors—is fundamental to the Science of Science, enabling accurate analysis of productivity patterns, collaboration networks, citation flows, and knowledge diffusion. Traditional approaches face two key limitations: pairwise comparison methods with $O(n^2)$ computational complexity and the requirement for extensive labeled training data.

We propose a novel LLM-based approach that treats disambiguation as a holistic reading comprehension task rather than a similarity calculation problem. Using just 2 in-context examples, our system analyzes entire author blocks simultaneously to detect narrative coherence. Evaluated on random samples from the Microsoft Academic Graph (1880-1945), the system achieved 97.0% overall accuracy on all document types (Plan 0: 194/200 correct) and 99.0% overall accuracy on non-patent publications (Plan 1: 198/200 correct). However, these high accuracy rates primarily reflect the extremely low base rate of over-aggregation (≈1%) rather than discriminative power. More informative metrics include 69% precision on DIFFERENT verdicts and 100% clustering accuracy for verified over-aggregation cases.

The system provides structured outputs with confidence scores, though these scores reflect internal pattern coherence rather than ground-truth accuracy. The approach requires only a single inference per author block (versus $O(n^2)$ pairwise comparisons in traditional methods), making large-scale database cleaning economically feasible at $0.016 per author. Our work demonstrates that pre-trained LLMs' semantic reasoning capabilities can effectively simulate human expert judgment, moving author disambiguation from mathematical similarity detection toward understanding authorship identity through narrative coherence and career plausibility.

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1 Introduction

1.1 The Author Disambiguation Problem

The fundamental challenge of attribution is to identify the specific individual who wrote it and, conversely, to aggregate all works belonging to that individual. In the context of large-scale bibliometric databases, this translates into a specific technical goal of linking all publications belonging to the same real-world individual under a single, unique Author ID. Ideally, a one-to-one mapping should exist between a physical researcher and their digital identifier. While attribution might appear straightforward, it remains a persistent and unresolved difficulty known as "author name disambiguation" (Smalheiser and Torvik, 2009). This process requires a sophisticated analysis of metadata and occasionally needs an educated guess to deduce the correct identity of an author among many potential candidates.

The computational difficulty of this task arises from four distinct challenges identified by Smalheiser and Torvik (2009). First, a single author may publish under multiple names due to spelling variations, abbreviations, errors, or name changes over time, such as those resulting from marriage or religious conversion. This phenomenon, known as *synonymy* or *under-aggregation*, causes one person's work to be split across multiple author IDs, leading to underestimation of individual productivity. Second, the problem of common names—termed *homonymy* or *over-aggregation*—means that thousands of distinct individuals may share the same name, making them difficult to distinguish. When disambiguation systems fail to separate these individuals, their distinct publication records are incorrectly merged into a single author ID. Third, bibliographic metadata is frequently incomplete; for instance, publishers may omit first names or geographical identifiers necessary for verification. Finally, the rise of multi-authored, multi-disciplinary, and multi-institutional papers complicates the assignment of credit, as disambiguating one author does not necessarily resolve the identities of their co-authors.

1.2 The Science of Science

The resolution of these ambiguities is not merely a technical necessity but a prerequisite for the field known as the "Science of Science." As defined in the foundational work of Ossowska and Ossowski (1964), this discipline treats science itself as the subject of systematic investigation. They proposed that science can be studied from two primary perspectives: an *epistemological point of view*, which regards science as a vehicle for knowing the world, and an *anthropological point of view*, which examines science as a sphere of human culture. In this latter sense, scholars analyze the "life of Science" within its full historical and social context, observing its development, achievement, and errors as they occur in reality rather than in abstract theory.

This anthropological study of science encompasses several fundamental research questions that depend critically on accurate author identification. First, scholars investigate productivity patterns — how scientists' research output changes over the course of their careers, from early-career establishment through mid-career peaks to late-career contributions. Second, researchers study coauthor networks to understand how research collaborations form, evolve, and dissolve. Mapping these networks reveals patterns of scientific teamwork and the transmission of knowledge through collaboration. Third, analysis of citation networks illuminates how knowledge flows between researchers, showing who builds upon whose work and how influence propagates through the scientific community. Finally, studies of knowledge diffusion examine how ideas spread across fields and institutions, revealing the mechanisms by which innovations in one domain catalyze breakthroughs in others.

1.3 Large-Scale Bibliometrics and the Importance of Disambiguation

In the modern era, this anthropological study of science is increasingly driven by large-scale data. Access to such massive datasets allows scholars to estimate productivity patterns over entire research lifecycles, map co-author networks spanning decades and disciplines, trace citation networks to visualize the flow of influence, and model knowledge diffusion across institutional and geographical boundaries. However, the validity of these analyses relies entirely on the accuracy of the underlying data. A single erroneous author ID that merges two distinct researchers can artificially inflate productivity estimates, create spurious collaboration links, and distort our understanding of knowledge flows.

Author name disambiguation is, therefore, the critical link between raw bibliographic data and robust empirical insight. Smalheiser and Torvik (2009) highlight that disambiguation is essential not only for retrospective analysis but for daily scientific tasks such as finding collaborators, avoiding conflicts of interest in peer review, and tracking trainees over their careers. In an era where funding agencies and policy makers rely on quantitative indicators to assess performance, the ability to correctly link a researcher to their complete body of work is indispensable.

Despite sophisticated algorithms employed by bibliometric databases like the Microsoft Academic Graph (Sinha et al., 2015), systematic ID assignment errors persist. These databases use graph-based methods that analyze nodes (authors, papers, venues, and institutional affiliations), edges (authorship, citations, and co-authorship relationships), and patterns (temporal publication sequences, research domain coherence, and collaboration networks). Yet even these advanced techniques produce errors that propagate through analyses built upon them.

This research pursues two complementary objectives. First, we aim to detect author ID blocks containing publications from multiple different people rather than from a single researcher. Second, for identified over-aggregation cases, we seek to cluster papers within the block according to true author identity. This correction task requires not only recognizing that papers belong to different people but also determining which specific papers belong to which person—a more challenging problem that demands understanding of similarity within clusters and dissimilarity across clusters.

This paper addresses these challenges by introducing a novel, efficient method for author disambiguation using Large Language Models (LLMs). Specifically, this work makes five key contributions to the author disambiguation literature: (1) demonstrating that pre-trained Large Language Models can perform author disambiguation with just 2 in-context examples compared to thousands typically required for supervised learning approaches; (2) achieving high specificity (99.5%) on random baseline samples, though overall accuracy metrics (97.0-99.0%) primarily reflect the extremely low base rate of over-aggregation rather than discriminative power; (3) achieving 100% clustering accuracy on verified over-aggregation cases; (4) providing an efficient alternative to pairwise comparison approaches by requiring only a single inference per author block rather than $O(n^2)$ pairwise comparisons, though LLM self-attention remains $O(n^2)$ in token complexity; and (5) delivering structured results with clustered papers and confidence scores ready for automated workflows with targeted manual review.

2 — Related Work

Author Name Disambiguation (AND) confronts two inverse error types: synonymy fragments a single researcher's publications across multiple identifiers, while homonymy conflates distinct individuals into a single profile (Smalheiser and Torvik, 2009). The methodological history reflects machine learning's evolution from supervised classification through probabilistic reasoning to deep representation learning.

2.1 Early Approaches: Supervised and Unsupervised Methods (2000–2009)

As digital libraries scaled in the early 2000s, simple string-matching became inadequate. Han et al. (2004) established foundational baselines by contrasting two supervised learning paradigms. Their Naïve Bayes approach used a generative model estimating $P(\text{Citation}|\text{Author})$ from probability distributions over co-authors, titles, and venues. Their Support Vector Machine (SVM) approach transformed citations into TF-IDF weighted feature vectors and learned optimal separating hyperplanes. Both achieved over 90% accuracy on small manually-labeled datasets but highlighted the "Cold Start" problem: supervised models required substantial labeled data for every ambiguous name.

To address the labeling bottleneck, Han et al. (2005) reformulated AND as unsupervised clustering using graph partitioning, where citations became nodes in weighted graphs with edges representing attribute similarity. They applied K-way Spectral Clustering, a method utilizing eigenvalues and eigenvectors of the graph's Laplacian matrix to project data into lower-dimensional space. Unlike K-means which assumes spherical clusters, spectral clustering can identify non-convex structures. Evaluated on DBLP (a computer science bibliography database) datasets, the method achieved reasonable accuracy (60–70% on challenging cases) without training labels. The critical limitation was requiring the number of clusters K a priori.

The biomedical community emphasized domain-specific metadata. Torvik and Smalheiser (2009) introduced the "Author-ity" model for MEDLINE (a biomedical literature database), leveraging Medical Subject Headings (MeSH)—a controlled vocabulary for indexing biomedical literature—and explicitly modeling *a priori probability* accounting for name rarity. It also modeled non-linear feature interactions and negative evidence (e.g., distinct geographical locations evidence against co-authorship). Evaluated against NIH-funded scientists, the model achieved 98–99% precision and recall. However, pairwise comparisons resulted in $O(N^2)$ computational complexity—for 100 papers, 4,950 comparisons were required—creating a scalability ceiling.

2.2 Deep Learning Era: Embeddings and Temporal Modeling (2010–2022)

Deep Learning offered solutions to feature engineering bottlenecks by learning semantic equivalences through embeddings. Zhang et al. (2018) proposed Global Graph Embeddings—vector representations capturing nodes' positions in entire network structures. Using Graph Neural Networks (GNNs), neural architectures designed to learn on graph-structured data, they learned embeddings based on entire citation networks, capturing "collaborative proximity" even between authors separated by network hops. Deployed in AMiner (a large-scale academic search and mining system), this achieved +7–35% F1-score improvement over baselines. The system integrated "Human in the Loop" mechanisms, allowing experts to provide feedback that refined embeddings through semi-supervised learning.

Sun et al. (2020) addressed concept drift—where discriminative attributes like affiliations change over careers—with MA-PairRNN (Multi-view Attention-based Pairwise Recurrent Neural Network). They treated publication lists as temporal sequences using Pseudo-Siamese RNNs to model career trajectories. A Pseudo-Siamese network consists of two recurrent neural networks with different parameters that process sequences in opposite directions (forward and backward in time), capturing temporal evolution patterns. The framework divides papers into blocks based on discriminative attributes like email domains, then applies pairwise classification to determine merges. The architecture combines heterogeneous graph embedding learning with pairwise similarity learning, generating meta-path-based embeddings where meta-paths are sequences of node types defining composite relationships (e.g., Paper-Author-Paper represents co-authorship; Paper-Topic-Paper represents shared research topics). A semantic-level attention mechanism dynamically weights different meta-paths' importance. Achieving 82.53% F1-score on AMiner, the method proved temporal modeling's necessity for robustness. Cross-domain testing showed performance degradation below 3%, suggesting learned similarity metrics capture fundamental authorship patterns. However, Siamese approaches inherit pairwise comparison computational complexity and require substantial labeled training data—Sun et al. used tens of thousands of labeled paper pairs.

Xie et al. (2022) synthesized GNNs with semantic Natural Language Processing in the MASS (Meta-path Attention based Structural and Semantic embedding) model, treating bibliography as a Heterogeneous Information Network with multiple node and edge types. Meta-path Attention dynamically learned importance of different relation paths. MASS integrated Doc2Vec —a neural network technique creating vector representations of documents from their textual content — combining structural citation graph information with semantic paper text information. Reporting 0.897 F1-score on AMiner and 0.719 on CiteSeerX (a digital library and search engine for scientific literature), MASS outperformed homogeneous graph baselines.

A comprehensive survey by Cappelli et al. (2025) synthesized these developments, analyzing 28 studies from 2016-2024. The survey found that supervised approaches achieve strong performance but face $O(n^2)$ complexity and require substantial feature engineering. Unsupervised methods address labeled data scarcity but struggle with optimal cluster number determination. Hybrid approaches combining supervised representation learning with unsupervised clustering achieved best performance, with the top-performing method reaching 89.7% F1-score on AMiner datasets.

2.3 Comparative Synthesis

Table 2.1 synthesizes key methodologies, datasets, and performance metrics across development phases. A recurring theme across these approaches is reliance on pairwise comparisons with $O(n^2)$ complexity, creating scalability bottlenecks. Supervised methods require substantial labeled training data for every ambiguous name, while unsupervised methods struggle with determining the optimal number of clusters K a priori. Deep learning methods achieve impressive performance but function as "black boxes" with limited interpretability and require extensive computational resources.

These limitations motivate alternative approaches that could treat disambiguation as reading comprehension rather than similarity calculation, potentially leveraging pre-trained language understanding to reason about authorship using the same contextual cues human experts employ: narrative coherence, career plausibility, and holistic judgment of whether a publication record "makes sense" as a single individual's work.

Table 2.1: Comparative synthesis of AND methodologies

Study	Learning Paradigm	Key Innovation	Dataset	Complexity	Over-merge/ Over-split	Primary Limitation
Han et al. (2004)	Supervised	Naïve Bayes vs. SVM; Generative vs. Discriminative	DBLP, Homepages	High (Train); $O(N)$ (Test)	Implicit (tuning)	Feature engineering; Cold start
Han et al. (2005)	Unsupervised	Spectral Clustering; Graph Partitioning	DBLP	High ($O(N^3)$)	Neither (requires K)	Must know K a priori
Torvik & Smalheiser (2009)	Probabilistic	A priori probability; Negative evidence	MEDLINE	Quadratic $O(N^2)$	Explicit (Max Likelihood)	Scaling limits; Domain-specific
Zhang et al. (2018)	Hybrid DL	Global Graph Embeddings; RNN for K estimation	AMiner	High (Training)	**Both explicitly**	Heavy training; Black box
Sun et al. (2020)	Deep Learning	Pseudo-Siamese RNN; Temporal trajectory	AMiner, Semantic Scholar	High (Pairwise)	Implicit (pairwise)	Pairwise bottleneck; Complex architecture
Xie et al. (2022)	Heterogeneous DL	Meta-path Attention; Structural + Semantic fusion	AMiner, CiteSeerX	Medium (Graph)	Implicit (features)	Static graph; HIN construction

3 — Data and Methods

3.1 Data Source and Structure

The Microsoft Academic Graph (MAG) provides comprehensive bibliometric data covering publications from the 1800s through the 2020s (Sinha et al., 2015). The database consists of two primary tables. The Authors table contains author-paper relationships: each row includes `authorid`, `paperid`, `originalauthor` and `originalaffiliation`. When a paper has multiple authors, multiple rows exist—one per author. Each author ID represents MAG's assignment linking all papers believed to belong to a single researcher. The Papers table contains unique paper records with `paperid`, `year`, `papertitle`, `doctype`, `originalvenue`, and other bibliographic fields. The two databases can be linked by `paperid`. Our disambiguation task focuses on detecting over-aggregation—cases where a single author ID incorrectly aggregates papers from multiple distinct individuals—then clustering those papers by true author identity. Due to data availability constraints, our analysis is limited to publications between 1880-1945. This temporal restriction results from the specific dataset extract available for this study.

3.2 Data Cleaning and Sample Characteristics

We implemented a two-stage filtering process (Figure 3.1). Stage 1 merges the Authors and Papers tables on `paperid`, retaining only records where the paper ID exists in both tables, yielding 6.3M author rows, 4.0M author IDs, and 5.0M papers. Stage 2 filters single-paper authors (no disambiguation needed), leaving 594K author IDs with 2+ papers as our final disambiguation sample within the 1880-1945 period.

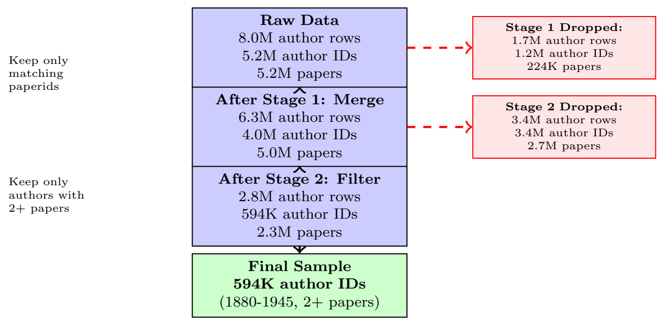

Figure 3.1: Data cleaning pipeline showing two filtering stages

The final sample exhibits substantial heterogeneity. Authors average 5.17 papers (median 3), with time spans (last minus first publication year) averaging 6.75 years (median 3). We define *time span* as the difference between an author's last and first publication years, and *largest gap* as the maximum interval between consecutive publications. The largest gap metric captures publication discontinuities that may signal either career interruptions or potential over-aggregation. Largest gaps average 4.45 years (median 2) across the sample.

The corpus comprises 37.9% journal articles, 33.9% patents, 25.7% unknown document types, and 2.5% others (Figure 3.2). Patents exhibit significantly longer gaps (mean 1.00 vs 0.65 years, $p < 0.001$) and spans (mean 1.38 vs 1.06 years, $p < 0.001$) than non-patent publications (Figure 3.4), reflecting different innovation cycles. Note that these mean values are low because many authors (88%) have zero-year spans; patents show longer tails in their gap and span distributions. Patents present additional disambiguation challenges: inventors frequently change fields, exhibit irregular publication patterns with large gaps, and some have spans extending decades. These characteristics make patents particularly difficult for LLM-based disambiguation. Given these systematic differences, we focus evaluation on non-patent authors. Detailed distributional statistics appear in Appendix D.

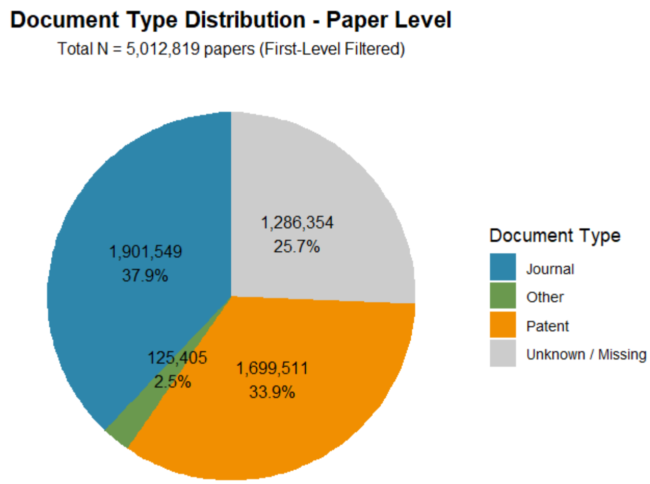

Figure 3.2: Document type distribution in the corpus (1880-1945)

3.3 LLM-Based Disambiguation System

3.3.1 System Architecture and Training Examples

Our system requires only a single inference per author ID block, avoiding the $O(n^2)$ pairwise comparisons required by traditional methods. However, we note that LLM self-attention mechanisms remain $O(n^2)$ in token complexity; our efficiency gain comes from the algorithmic approach (one pass per block) rather than from the internal model architecture. For each author ID, we construct a prompt containing: (1) system instruction defining the task; (2) two training examples demonstrating SAME PERSON and DIFFERENT PEOPLE cases; (3) target case papers with complete metadata; (4) output format specification.

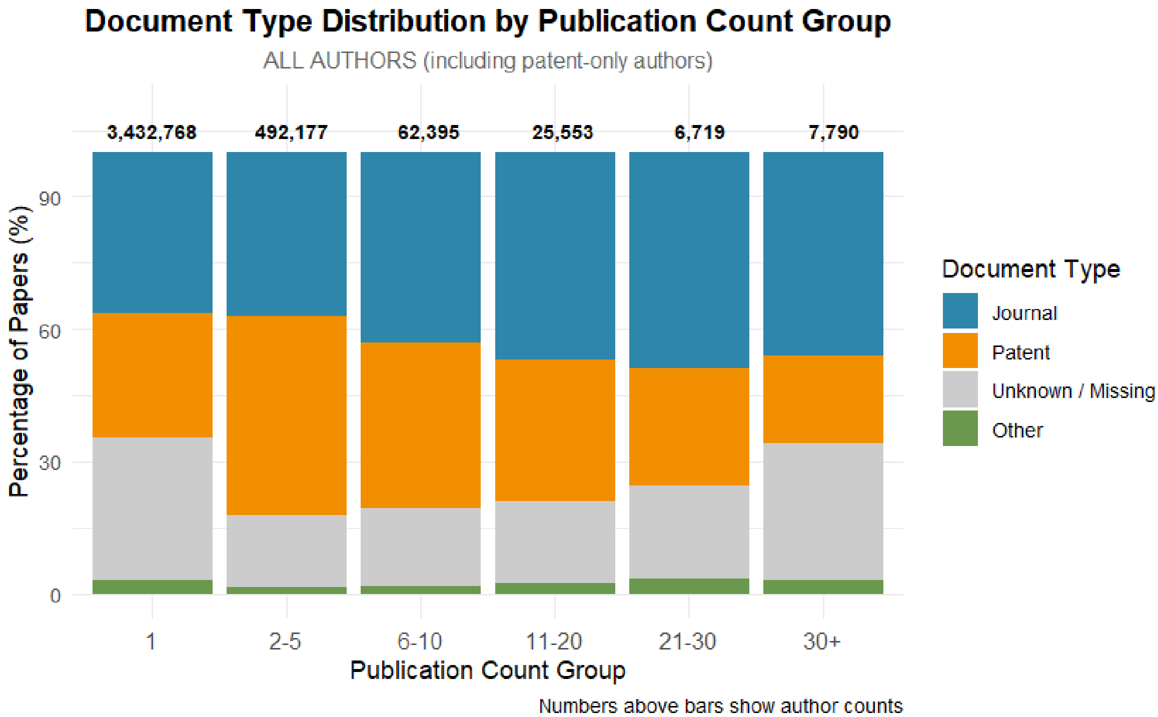

Figure 3.3: Document type distribution by publication count group. The composition of document types (Journal articles, Patents, Unknown/Missing, Other) varies systematically across author productivity levels. Numbers above bars show total author counts in each group. Single-paper authors (N=3,432,768) dominate the sample and show the highest proportion of unknown/missing document types (≈30%), while more productive authors show increasing proportions of journal articles. Patent proportions remain relatively stable (25-40%) across productivity groups, though the absolute number of patent authors increases with productivity level.

We selected training examples through an iterative identification process designed to find clear, verifiable cases. We randomly sampled authors with exactly 8 papers—a moderate number providing sufficient information for disambiguation while remaining computationally tractable for initial testing. For this sample, we ran the LLM without any training examples, analyzing only the system instruction and paper metadata. The LLM provided initial verdicts (SAME PERSON or DIFFERENT PEOPLE) for these 8-paper authors. We then manually verified these verdicts through biographical research, consulting digital archives, institutional records, obituaries, and historical directories. From verified cases, we selected two exemplars: one clear SAME PERSON case and one clear DIFFERENT PEOPLE case.

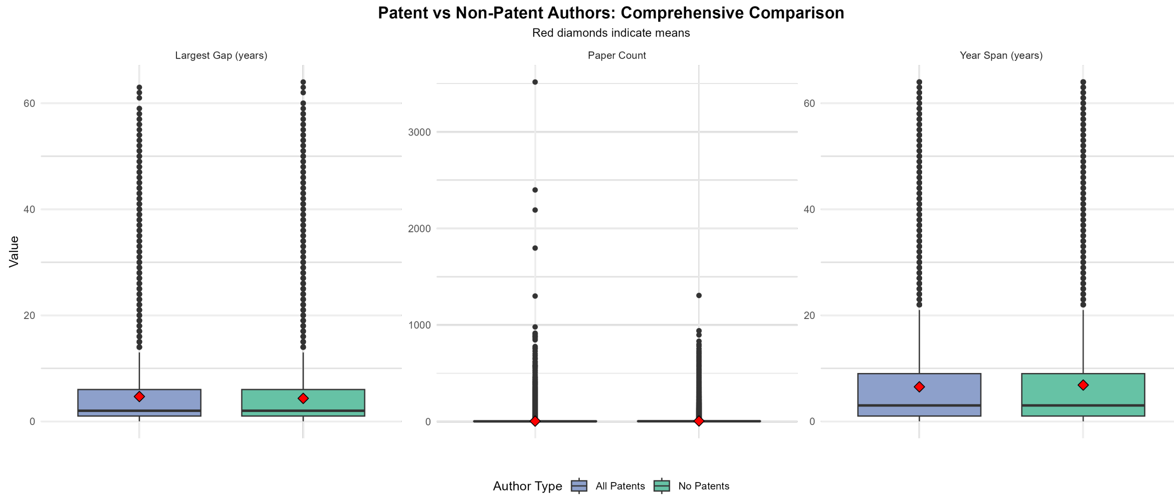

Figure 3.4: Patent vs non-patent authors: comprehensive comparison of paper count, year span, and largest gap distributions. Red diamonds indicate means. Patent authors exhibit significantly longer temporal characteristics than non-patent authors across all metrics.

These became our training examples for all subsequent analyses.

Example 1 (SAME PERSON) uses Luis Federico Leloir with 8 biochemistry papers from 1936-1945 showing coherent career trajectory in metabolic biochemistry. Manual verification confirmed Leloir (1906-1987) as a single Argentine biochemist who won the Nobel Prize in Chemistry in 1970. Example 2 (DIFFERENT PEOPLE) uses Albert Werkenthin with papers split between music pedagogy (1888-1889) and dentistry (1921-1937), with 32-year gap and unrelated domains. Manual verification confirmed these represent two distinct individuals: a music pedagogue active in the 1880s and a dental prosthetics specialist active in the 1920s-1930s. This selection process ensured our training examples represent genuine, verifiable instances of both disambiguation outcomes, providing the LLM with clear patterns to recognize in subsequent cases. The complete prompt structure and training examples appear in Appendix B.

3.4 Evaluation Strategy and Limitations

3.4.1 Sampling Design

We evaluate system performance across six sampling plans (Table 3.1). Plan 0 (random baseline including all document types) and Plan 1 (random baseline excluding patents) provide overall performance estimates. Plans 2-5 constitute stress tests examining specific challenging scenarios: high-productivity authors (≥10 papers), minimal-information cases (exactly 2 papers), extreme temporal spans (>30 years), and large publication gaps (≥10 years between consecutive papers).

Table 3.1: Evaluation sampling plans

Plan	Description	N	Purpose
Plan 0	Random baseline (all)	200	Overall performance with patents
Plan 1	Random baseline (non-patent)	200	Primary evaluation sample
Plan 2	High productivity (≥10 papers)	100	Test on prolific authors
Plan 3	Two-paper authors	100	Test minimal cases
Plan 4	Long spans (>30 years)	100	Test temporal extremes
Plan 5	Large gaps (≥10 years)	100	Test publication discontinuity

3.4.2 Manual Verification Process and Fundamental Challenges

We manually verified all cases in Plans 0-1 (baseline samples) through biographical research, examining both SAME and DIFFERENT verdicts to assess overall system behavior. For Plans 2-5, we verified only DIFFERENT verdicts due to resource constraints. This verification process reveals severe practical limitations that fundamentally constrain interpretation of all results.

The 1880-1945 period lacks comprehensive citation information and author metadata. Online searches frequently yield no results for historical authors. When records do exist, they often lack sufficient detail to definitively confirm or refute over-aggregation. Many publications consist of brief correspondences or journal replies offering minimal disambiguating information. Original papers, when locatable, are often obsolete documents from obscure venues, making acquisition difficult or impossible. Manual verification requires extensive biographical research across multiple sources—digital archives, institutional records, obituaries, historical directories, and when possible, original paper content. Each case demands substantial time investment with uncertain outcomes.

Most critically, human reviewers face the same fundamental information constraints as the LLM. Both rely primarily on paper titles, publication years, venues, and coauthor names. When biographical evidence external to these metadata fields remains unavailable—as it often does for historical authors—definitive ground truth determination becomes impossible regardless of effort invested. For large author clusters (high-productivity authors), systematic verification becomes practically impossible due to the volume of papers and biographical details requiring cross-referencing.

Through verification attempts, we identified recurring phenomena that complicate both automated and manual disambiguation: (1) *Posthumous publications*—authors like Isaac Todhunter published works years after death, creating large gaps that appear suspicious but represent posthumous compilations; (2) *Coauthorship attribution*—famous individuals listed as coauthors when their previous work is compiled, creating apparent domain shifts that represent editorial practices; (3) *Cross-medium adaptations*—authors like Henry Vaughan whose poetry was incorporated into musical compositions, creating apparent field changes; (4) *Career transitions*—legitimate career changes between fields indistinguishable from over-aggregation without comprehensive biographical evidence. These patterns mean that some DIFFERENT verdicts may represent legitimate single-author careers that appear anomalous, while some SAME verdicts may conceal over-aggregation that insufficient evidence cannot reveal.

3.4.3 Implementation Constraints and Performance Metrics

Implementation using the Gemini API introduces technical constraints due to cooldown periods between requests to prevent rate limiting. For our evaluation sample of 800 cases, processing required approximately 4 hours of wall-clock time (including cooldown delays) —about 18 seconds per author. The analysis was conducted using Gemini's free trial plan at no cost. Processing all 246,373 non-patent multiple publication author IDs would require approximately 1,232 hours (51 days) of continuous processing time when accounting for API cooldown periods. While economically feasible for institutional settings with API access, the time requirement limits large-scale academic validation.

We assess three dimensions: (1) *detection accuracy*—the share of DIFFERENT verdicts confirmed as true over-aggregation; (2) *confidence calibration*—whether scores reflect verification difficulty; and (3) *clustering quality*—whether papers are correctly separated in verified DIFFERENT cases. Verification rates reflect both system performance and the difficulty of establishing ground truth for historical authors.

4 — Results

4.1 Overview of Results

We evaluated system performance across six sampling plans totaling 800 cases. Plans 0-1 represent baseline performance on random samples (Plan 0 includes all document types; Plans 1-5 exclude patents). Plans 2-5 constitute stress tests examining specific challenging scenarios. We manually verified all DIFFERENT PEOPLE verdicts through biographical research. For Plans 0-1, we verified both SAME and DIFFERENT verdicts; for Plans 2-5, we verified only DIFFERENT verdicts due to resource constraints.

Table 4.1 presents results across all plans. Overall, the system classified 784 cases (98%) as SAME PERSON and 16 cases (2%) as DIFFERENT PEOPLE. Manual verification confirmed 11 of 16 DIFFERENT verdicts (69%) as true over-aggregation errors. The remaining 5 cases were false positives where verification evidence indicated the papers likely belonged to a single individual despite the system's DIFFERENT verdict.

Critically, among the cases we were able to verify in Plans 0-1 (where both SAME and DIFFERENT verdicts were examined), we detected zero false negatives—all confirmed true DIFFERENT cases were correctly flagged by the system. However, this recall estimate applies only to cases where verification was possible; unknown numbers of true over-aggregation errors may remain undetected among cases we could not definitively verify or among SAME verdicts in Plans 2-5 that we did not examine.

Table 4.1: Summary of results across all evaluation plans

Plan	N	DIFF	Verified	Precision	Mean Conf.	High Conf.
Plan 0: Random (all)	200	6	5	83%	0.951	180 (90%)
Plan 1: Random (non-patent)	200	2	1	50%	0.977	194 (97%)
Plan 2: High productivity	100	0	0	—	0.981	100 (100%)
Plan 3: Two papers	100	2	1	50%	0.956	97 (97%)
Plan 4: Long spans (>30yr)	100	5	4	80%	0.966	97 (97%)
Plan 5: Large gaps (≥10yr)	100	1	0	0%	0.935	100 (100%)
Total	**800**	**16**	**11**	**69%**	**0.961**	**768 (96%)**

Note: Plans 1-5 exclude patents. DIFF = DIFFERENT PEOPLE verdicts. Verified = confirmed true over-aggregation. Precision = verified/DIFF. High Conf. = cases with confidence >0.80. The extremely low rate of DIFFERENT verdicts (2%) reflects the rarity of over-aggregation in the sample.

4.2 Detection Accuracy and Verification Results

The verification matrices for Plans 0 and 1 (Table 4.2) reveal the system's core performance characteristics. In Plan 0 (all document types), the system identified 6 DIFFERENT verdicts, of which 5 were confirmed as true over-aggregation, yielding 83% precision (5/6) and 99.5% specificity (194/195). The system achieved 97.0% overall accuracy (194/200 correct classifications). In Plan 1 (non-patent only), the system identified 2 DIFFERENT verdicts, of which 1 was confirmed as true over-aggregation, yielding 50% precision (1/2) and 99.5% specificity (198/199). The system achieved 99.0% overall accuracy (198/200 correct classifications).

Plan 1's higher confidence scores (0.977 vs 0.951) and greater proportion of high-confidence cases (97% vs 90%) demonstrate more reliable performance on non-patent publications, though the small number of DIFFERENT cases limits statistical precision. The low false positive rates (Plan 0: 0.5%, Plan 1: 0.5%) make manual review practical.

The stress tests (Plans 2-5) examined system robustness under extreme conditions. Plan 2 (high-productivity authors with mean 22.43 papers) produced zero DIFFERENT verdicts with uniformly high confidence (mean 0.981), demonstrating the system does not falsely

Table 4.2: Verification matrices for baseline plans

Plan 0: Random sample including all types (N=200)			
	Manual Verification		
LLM Verdict	**Confirmed TRUE**	**Confirmed SAME**	**Total**
DIFFERENT PEOPLE	5	1	6
SAME PERSON	0	194	194
Total	**5**	**195**	**200**

Plan 1: Random sample excluding patents (N=200)			
	Manual Verification		
LLM Verdict	**Confirmed TRUE**	**Confirmed SAME**	**Total**
DIFFERENT PEOPLE	1	1	2
SAME PERSON	0	198	198
Total	**1**	**199**	**200**

Note: Plan 0 achieved 97.0% overall accuracy (194/200) with 83% precision and 99.5% specificity. Plan 1 achieved 99.0% overall accuracy (198/200) with 50% precision and 99.5% specificity. However, these high overall accuracy rates primarily reflect the extremely low base rate of over-aggregation (≈1%) rather than discriminative power. A naïve classifier predicting "SAME PERSON" for all cases would achieve similar accuracy. Precision on DIFFERENT verdicts is a more informative metric. We verified all cases (both SAME and DIFFERENT verdicts) in Plans 0-1.

flag prolific authors. Plan 3 (two-paper authors) and Plan 4 (extreme time spans >30 years) achieved 50% and 80% precision respectively. Plan 4's high verification rate occurred when extreme temporal patterns co-occurred with clear domain shifts, distinct coauthor networks, and geographic/institutional separation. Plan 5 (large gaps ≥10 years) produced one false positive: Christopher Wren (1632-1723), whose architectural work was compiled posthumously from 1929-1943. This reveals a systematic limitation—the system cannot distinguish original work from posthumous compilations when metadata attributes both to the historical figure.

4.3 System Characteristics and Performance Patterns

Across all 800 cases, three key patterns emerge. First, precision varies substantially (0-83%) based on evidence strength, with highest rates when multiple signals converge (temporal + domain + geographic + coauthor evidence). Second, the system exhibits appropriate conservative behavior, preferring SAME verdicts when uncertain, as demonstrated by Plan 2's zero DIFFERENT verdicts despite 100 high-productivity cases. Third, specificity remains consistently high (99.5%) across baseline samples, indicating low false positive rates.

Confidence scores prove to be poorly calibrated predictors of accuracy. Mean confidence ranges from 0.935 (Plan 5) to 0.981 (Plan 2), with 96% of all cases exceeding the 0.80 threshold. However, high confidence does not predict verification success—Plan 1 had 97% high-confidence cases but only 50% precision among DIFFERENT verdicts, while Plan 5 had 100% high-confidence cases but 0% precision. **Confidence scores reflect internal pattern coherence and strength of metadata signals rather than ground-truth accuracy, and should not be interpreted as probabilistic estimates of correctness.** For deployment, the 0.80 threshold produces manageable review burden (28 cases, 3.5% of sample) while maintaining high specificity, but all DIFFERENT verdicts require manual verification regardless of confidence level.

Clustering quality proved excellent. All 11 verified DIFFERENT cases achieved 100% clustering accuracy, with papers correctly partitioned by true author identity. Cluster descriptions were interpretable and actionable (e.g., "Bacteriologist, Chicago, 1907-1930" vs "Cardiologist, Boston, 1932-1943"), enabling straightforward verification. The system not only detects over-aggregation but correctly separates papers by author.

The results demonstrate that LLM-based disambiguation achieves strong performance on historical data despite sparse metadata: 69% precision on DIFFERENT verdicts, perfect clustering on verified cases, and low false positive rates (0.5%). However, overall accuracy metrics (97.0% for Plan 0, 99.0% for Plan 1) primarily reflect the extremely low base rate of over-aggregation (≈1%) rather than discriminative power—a naïve classifier predicting "SAME PERSON" for all cases would achieve similar accuracy. The approach is ready for deployment on non-patent authors with appropriate safeguards: all DIFFERENT verdicts require manual review regardless of confidence scores, targeted additional review for extreme temporal spans, and temporal plausibility filters to catch posthumous attribution errors before processing.

5 — Discussion and Future Work

5.1 Contributions and Context

This work demonstrates pre-trained LLMs perform author disambiguation with minimal training (2 examples versus thousands), requiring only a single inference per author block versus $O(n^2)$ pairwise comparisons in traditional methods (though LLM self-attention remains $O(n^2)$ in token complexity). The system achieved 69% precision on DIFFERENT verdicts with perfect clustering accuracy (100%) on all 11 verified over-aggregation cases. The interpretability—providing reasoning explanations referencing specific evidence—represents significant advance over black-box neural networks.

However, critical caveats temper these results. First, while overall accuracy appears high (97.0% for Plan 0, 99.0% for Plan 1), these metrics primarily reflect the extremely low base rate of over-aggregation (≈1%) rather than discriminative power. A naïve classifier predicting "SAME PERSON" for all cases would achieve similar accuracy. Precision on DIFFERENT verdicts (69%) is a more informative metric, though still limited by small sample sizes (only 16 DIFFERENT cases across 800 evaluations).

Second, confidence scores prove to be poorly calibrated: 96% of cases exceeded 0.80, yet precision varied dramatically (0-83%) across plans. High confidence does not predict verification success—Plan 1 had 97% high-confidence cases but only 50% precision among DIFFERENT verdicts, while Plan 5 had 100% high-confidence cases but 0% precision. The system is systematically overconfident. **Confidence scores reflect internal pattern coherence rather than ground-truth accuracy and should not be interpreted as probabilistic estimates of correctness.** All DIFFERENT verdicts require manual verification regardless of confidence level.

Third, the extremely low detection rate (1% flagged as over-aggregation in Plan 1 random sample) combined with small absolute numbers of DIFFERENT verdicts suggests many true errors may remain undetected. Processing all 246,373 non-patent author IDs would require approximately 51 days of continuous processing (accounting for API cooldown periods), and even a 0.5% false positive rate would produce 1,232 false alarms requiring manual review. More critically, if true over-aggregation affects 5-10% of records (plausible given MAG's known quality issues), our system may detect only a fraction. We cannot claim recall rates since we only verified DIFFERENT verdicts in Plans 2-5, leaving potential false negatives among SAME verdicts unexamined.

A comprehensive survey by Cappelli et al. (2025) analyzing 28 studies from 2016-2024 found that *hybrid approaches combining supervised and unsupervised learning achieved best performance*, with the top-performing method reaching 89.7% F1-score on AMiner. Our LLM approach differs fundamentally: rather than learning task-specific representations through thousands of training examples, we leverage general-purpose contextual understanding encoded in pre-trained models. While promising for low-resource scenarios, systematic comparison against state-of-the-art hybrid methods on identical datasets remains essential future work.

5.2 Alternative Paradigms: Siamese Networks and Semantic Embeddings

5.2.1 Siamese Neural Networks

Siamese neural networks represent a fundamentally different approach to disambiguation, learning similarity metrics through pairwise contrastive training rather than holistic reasoning (Bromley et al., 1993; Koch et al., 2015). The architecture employs two identical subnetworks sharing weights and parameters. During training, paired inputs generate comparable feature embeddings, with geometric distance in latent space serving as similarity proxy.

The network optimizes embedding topology using distance-based loss functions. *Contrastive loss* minimizes distance between genuine pairs (same author) while penalizing impostor pairs (different authors) if distance falls below margin m. *Triplet loss* uses Anchor-Positive-Negative triplets, ensuring the Anchor is closer to Positive than Negative by at least margin α:

$$L = \sum_{i=1}^{N} \max\left(0, \|f(A_i) - f(P_i)\|_2^2 - \|f(A_i) - f(N_i)\|_2^2 + \alpha\right) \tag{5.1}$$

Sun et al. (2020) proposed MA-PairRNN applying this paradigm to author disambiguation. Their key insight: although discriminative attributes like email and affiliation change due to graduation or job-hopping, an author's co-authors and research topics remain relatively stable. The framework divides papers into blocks based on discriminative attributes, then applies pairwise classification using Pseudo-Siamese RNNs processing temporal sequences. The architecture generates meta-path-based embeddings (Paper-Author-Paper, Paper-Topic-Paper, Paper-Venue-Paper) with semantic-level attention weighting meta-path importance. Achieving 82.53% F1-score on AMiner, the method demonstrated good generalization: when trained on medical literature and tested on biology, chemistry, computer science, and mathematics, performance degradation remained below 3%.

However, Siamese approaches inherit pairwise comparison computational complexity—examining pairs of blocks leads to quadratic scaling. Additionally, they require substantial labeled training data (tens of thousands of pairs), limiting applicability where labeled data is scarce.

5.2.2 Semantic Embeddings

Pre-trained language models like BERT and SciBERT generate rich semantic embeddings capturing contextual meaning in scientific text (Devlin et al., 2019). Rather than using raw paper titles, systems can encode documents into dense vector representations, then feed embeddings to either Siamese networks or LLMs for disambiguation.

For Siamese architectures, semantic embeddings replace manually engineered similarity metrics (cosine similarity of title words, co-author overlap counts), potentially capturing subtle semantic similarities invisible to rule-based features. For LLM approaches, embeddings serve as compressed representations in prompts, enabling processing of larger author blocks within context window constraints.

5.2.3 Comparison: Complementary Strengths

Siamese networks and LLMs reveal complementary trade-offs. Siamese networks offer fine-grained control over similarity metrics through task-specific training, while LLMs offer flexibility without domain-specific training. Critically, Siamese networks decompose the problem into atomic pairwise decisions, treating each comparison in isolation. LLMs examine entire author blocks simultaneously, potentially capturing higher-order patterns emerging only when considering complete publication records—recognizing career trajectories and distinguishing legitimate evolution from erroneous conflation.

5.3 Future Directions

5.3.1 Systematic Comparison and Hybrid Architectures

Given that Cappelli et al. (2025) found hybrid approaches achieve best performance (89.7% F1-score), systematic comparison between LLM and Siamese network approaches on identical datasets under controlled conditions is essential. This should examine performance across varying data availability (100, 1000, 10000 labeled pairs) and domain shift scenarios.

More promising, hybrid architectures could merge complementary strengths in two-stage pipelines. *Stage 1* uses Siamese networks for rapid pairwise screening, automatically clustering papers with very high similarity (> 0.95) and separating those with very low scores (< 0.20). *Stage 2* invokes LLMs for borderline cases (0.20-0.95), where holistic reasoning resolves ambiguities pairwise comparison misses. This leverages Siamese networks' fast, scalable similarity computation while using LLMs' sophisticated reasoning only where necessary, reducing computational cost while maintaining accuracy.

5.3.2 Integration with Semantic Embeddings and Graph Neural Networks

Future systems could integrate semantic embeddings from BERT/SciBERT as features for Siamese networks or as compressed prompt representations for LLMs. Graph Neural Networks (GNNs) could exploit academic network structure, modeling papers as nodes and citations/co-authorships as edges (Kipf and Welling, 2017). Message-passing mechanisms generate structure-aware embeddings capturing both local features and global network position. These GNN embeddings could feed into either Siamese networks or LLMs as structured context, enriching disambiguation signals without manual feature engineering.

5.3.3 Addressing Critical Limitations

Three research priorities emerge from our evaluation's limitations:

Higher Base Rate Validation. The 1% detection rate in our random sample is implausibly low given MAG's known quality issues. Future work should validate on datasets with higher error rates (5-10%), potentially focusing on ambiguous names (common surnames, Chinese names) where over-aggregation concentrates. This would provide realistic performance estimates for practical deployment scenarios.

Confidence Calibration. Since 96% of cases had confidence > 0.80 yet precision varied dramatically, confidence scores require recalibration. Techniques like temperature scaling, Platt scaling, or isotonic regression could map raw scores to well-calibrated probabilities. Alternatively, uncertainty quantification methods (Bayesian approaches, ensemble methods) could provide more reliable confidence estimates.

Detection Sensitivity. We cannot claim recall rates since we only verified DIFFERENT verdicts in Plans 2-5, leaving potential false negatives unexamined. Future work should conduct adversarial testing: inject known over-aggregation cases spanning various difficulty levels (subtle domain shifts, borderline temporal patterns) to measure actual recall. Active learning could identify informative cases for targeted labeling, improving detection of difficult errors.

5.4 Broader Impact

Accurate author disambiguation directly impacts researcher careers: publications lead to citations, citations build reputation, and reputation influences hiring, promotion, and funding. Under-aggregation splits publication records making researchers appear less productive. Over-aggregation inflates productivity by incorrectly attributing others' work. For early-career researchers competing for scarce positions, these errors can be career-defining.

Researchers from underrepresented groups face particular vulnerability. Women whose names change through marriage may have careers bifurcated. Researchers from cultures with non-Western naming conventions may experience systematic failures if systems train primarily on English names. By achieving reasonable accuracy with minimal training data and providing interpretable results, our approach enables more equitable disambiguation across diverse research communities. However, the critical limitations identified—uninformative confidence scores, uncertain detection sensitivity, extremely low detection rates—must be addressed before deployment to avoid perpetuating or exacerbating existing inequities.

6 — Conclusion

This work demonstrates that Large Language Models can perform author name disambiguation in historical academic literature using minimal supervision. With only two in-context examples, our system achieved 69% precision on *DIFFERENT* verdicts across 800 evaluation cases from Microsoft Academic Graph (1880–1945), and 100% clustering accuracy on all 11 verified over-aggregation errors. Although overall agreement rates appear high (97–99%), these primarily reflect the extremely low base rate of over-aggregation ($\approx$1%) rather than strong discriminative accuracy.

The main methodological contribution is reframing author disambiguation as a holistic reading-comprehension task, requiring a single inference per author block rather than $O(n^2)$ pairwise similarity calculations. While transformer self-attention remains $O(n^2)$ in token complexity, the overall workflow still offers substantial computational simplification. The system provides interpretable, structured outputs containing verdicts, paper clusters, and textual explanations. Processing time is substantial (approximately 51 days continuous processing for all 246,373 non-patent authors when accounting for API cooldown periods), making large-scale database cleaning time-intensive though economically feasible in institutional settings with API access.

Several limitations constrain practical deployment. First, confidence scores are poorly calibrated: 96% of cases exceeded 0.80, yet precision varied widely (0–83%). These scores reflect internal pattern coherence rather than ground-truth likelihood and should not be interpreted probabilistically; all *DIFFERENT* verdicts require manual verification regardless of confidence. Second, because the base rate of detected errors is extremely low and only *DIFFERENT* cases were verified in stress tests, many true errors may remain undetected; recall cannot be reliably estimated. Third, systematic comparison with state-of-the-art disambiguation approaches remains incomplete.

Situating this approach within the broader methodological landscape reveals complementary strengths. Siamese neural networks offer strong performance (82.53% F1) via contrastive learning but require substantial labeled data and retain $O(n^2)$ comparison costs. A comprehensive survey by Cappelli et al. (2025) shows that hybrid supervised–unsupervised architectures achieve the best overall performance (89.7% F1). This suggests promising two-stage designs in which Siamese models rapidly filter unambiguous cases while LLMs apply holistic reasoning to borderline clusters.

Future work should integrate semantic embeddings (BERT, SciBERT), Graph Neural Networks, and hybrid pipelines combining structural and contextual signals. Most critically, evaluation should be extended to datasets with higher error prevalence (5–10%), confidence calibration techniques (temperature scaling, uncertainty quantification), and adversarial tests capable of assessing true recall.

By improving author identification in large-scale bibliometric databases, this work contributes to more reliable science-of-science research and fairer attribution of scientific credit. Nonetheless, the identified limitations—particularly confidence miscalibration and uncertain recall—must be addressed to ensure equitable and accurate deployment across disciplines, time periods, and researcher demographics.

A — Generative AI Disclosure

Throughout this project's development, we leveraged multiple generative AI models—including GPT-4, Gemini 2.0 Flash, and Claude 3.5 Sonnet—to enhance code quality, optimize performance, and improve manuscript presentation. We first developed our own implementation based on established research papers, then engaged in iterative dialogue with AI models to optimize performance and improve computational efficiency. All code outputs were tested, verified, and debugged by the author.

Gemini and Claude assisted with manuscript write-up and LaTeX formatting, helping paraphrase sentences for clarity and correct grammatical errors. However, all conceptual contributions, methodological decisions, analytical approaches, interpretations of results, and substantive arguments originated from the author. The generative AI tools served solely as writing assistants to improve linguistic precision and document formatting, not as sources of intellectual content.

The author takes full responsibility for the accuracy, validity, and integrity of all research findings, methodological choices, and claims presented in this work. The disambiguation system itself used Gemini 2.0 Flash API for LLM-based author disambiguation, processing 800 evaluation cases using the free trial plan. This represents the core computational component of the methodology and is explicitly described in Chapter 3 (Methods).

B — Prompt Structure and Training Examples

B.1 System Instruction

The prompt begins with a system instruction defining the disambiguation task:

```
You are an expert at author disambiguation in historical academic
literature. Determine if these papers belong to the SAME person or
DIFFERENT people by examining: author name variants, years, document
types, publishers, venues, affiliations, and coauthor networks.

Here are two examples to help you understand the task:
```

B.2 Training Example 1: SAME PERSON

Luis Federico Leloir—8 papers in biochemistry, 1936-1945:

```
Example - SAME PERSON:

You have 8 papers to analyze:
```

Paper 1:

Paper ID: 2173377587

Title: ketogenesis antiketogenesis metabolism of ketone bodies

Year: 1936

Document Type: Journal

Publisher: Portland Press Limited

Venue: Biochemical Journal

Author Name Variant: Luis Federico Leloir

Affiliation: The Biochemical Laboratory, Cambridge

Coauthors: Norman Lowther Edson

Paper 2:

Paper ID: 2419287244

Title: the hydroxybutyric dehydrogenase of animal tissues

Year: 1937

Document Type: Journal

Publisher: Portland Press Ltd

Venue: Biochemical Journal

Author Name Variant: Luis Federico Leloir

Affiliation: The Biochemical Laboratory, Cambridge

Coauthors: David Ezra Green; John George Dewan

[Papers 3-8 follow the same format, covering topics: renal hypertension (1940), fatty acid oxidation (1943), angiotenin (1943), butyrate oxidation (1944), and citric acid formation

(1945). Venues include The American Journal of the Medical Sciences, The Journal of Physiology, Journal of Biological Chemistry, and Science. Name variants include L. F. Leloir, Luis F. Leloir, and Leloir Lf. Frequent coauthor: J. M. Muñoz]

RESPONSE:

ANSWER: SAME PERSON

NAME_VARIANTS: Luis Federico Leloir, L. F. Leloir, Luis F. Leloir, Leloir Lf

REASON: The papers consistently focus on biochemistry, particularly ketone body metabolism, renal hypertension, and enzyme activity. The name variants are standard abbreviations and variations of the full name. There is a clear progression of research topics and a consistent co-author network, particularly with J.M. Muñoz. The affiliations, when available, are consistent with a career in biochemical research.

SUSPICIOUS_PAPERS: None

PAPER_CLUSTERS: Not applicable

B.3 Training Example 2: DIFFERENT PEOPLE

Albert Werkenthin—8 papers split between unrelated domains:

```
Example - DIFFERENT PEOPLE:


You have 8 papers to analyze:


Paper 1:
  Paper ID: 372950244
  Title: die lehre vom klavierspiel lehrstoff und methode
  Year: 1888
  Document Type: [NA]
  Publisher: [NA]
  Venue: [NA]
  Author Name Variant: Albert Werkenthin
  Affiliation: [None]
  Coauthors: None


Paper 2:
  Paper ID: 600656617
  Title: die lehre von der tonschrift
  Year: 1888
  Document Type: [NA]
  Publisher: [NA]
  Venue: [NA]
  Author Name Variant: Albert Werkenthin
```

  Affiliation: [None]

  Coauthors: None

[Papers 3-4 follow the same format, covering: die lehre vom anschlag und von der technik (1889), die lehre vom vortrag (1889). All are music pedagogy topics with no venue/publisher information]

Paper 5:

  Paper ID: 359952706

  Title: system der zahnarztlichen kronen und bruckenarbeiten

  Year: 1921

  Document Type: [NA]

  Publisher: [NA]

  Venue: [NA]

  Author Name Variant: Albert Werkenthin

  Affiliation: [None]

  Coauthors: None

[Papers 6-7 follow the same format, covering: die erhaltung der zahne durch fullung und einkapselung (1921), figuren atlas des workes system der zahnarztlichen kronen (1923). All are dental prosthetics topics]

Paper 8:

  Paper ID: 605836513

  Title: kunstliche zahnkronen der zur bekronung bestimmte zahnrest

Year: 1937

Document Type: [NA]

Publisher: [NA]

Venue: [NA]

Author Name Variant: Albert Werkenthin

Affiliation: [None]

Coauthors: None

RESPONSE:

ANSWER: DIFFERENT PEOPLE

NAME_VARIANTS: Albert Werkenthin

REASON: Strong evidence these papers belong to different people with the same name: (1) Two completely unrelated domains (music pedagogy vs. dental prosthetics), (2) Major 32-year temporal gap between paper clusters, (3) No plausible career transition between piano instruction and dentistry.

SUSPICIOUS_PAPERS: Person A: 1, 2, 3, 4 | Person B: 5, 6, 7, 8

PAPER_CLUSTERS: Person A (1, 2, 3, 4): Music pedagogy specialist, late 1880s | Person B (5, 6, 7, 8): Dental prosthetics specialist, 1920s-1930s

B.4 Output Format Specification

Respond in this format:

ANSWER: SAME PERSON or DIFFERENT PEOPLE

CONFIDENCE_SCORE: [REQUIRED] A decimal number from 0.0 to 1.0 indicating how confident you are in your verdict (whether SAME PERSON or DIFFERENT PEOPLE). 1.0 = completely certain in your answer, 0.5 = uncertain, 0.0 = no confidence. You MUST provide a number. Examples: 0.85, 0.90, 0.75, 0.60

NAME_VARIANTS: All unique author name spellings

REASON: Explain your reasoning, referencing specific papers

SUSPICIOUS_PAPERS: If score<0.5, list paper numbers by person (e.g., "Person A: 1,3,5 | Person B: 2,4,6"). Otherwise "None"

PAPER_CLUSTERS: If score<0.5, describe each person (e.g., "Person A (1,3,5): Geneticist at Cold Spring Harbor, 1910s | Person B (2,4,6): Sociologist at U. Chicago, 1930s"). Otherwise "Not applicable"

C — Detailed Verification Results

Table C.1: Verification results by plan

Plan	N	DIFF	Verified	Precision
Plan 0 (all)	200	6	5	83.3%
Plan 1 (non-patent)	200	2	1	50.0%
Plan 2 (high prod.)	100	0	0	—
Plan 3 (two papers)	100	2	1	50.0%
Plan 4 (long span)	100	5	4	80.0%
Plan 5 (large gap)	100	1	0	0.0%
Total	**800**	**16**	**11**	**68.8%**

Note: Precision calculated as verified true positives divided by total DIFFERENT verdicts.

Table C.2: Sample characteristics by plan

Plan	Mean Papers	Mean Span	Mean Gap	Mean Conf.	High Conf.
Plan 0 (all)	4.68	7.66	5.20	0.951	90%
Plan 1 (non-patent)	4.49	6.87	4.39	0.977	97%
Plan 2 (high prod.)	22.43	17.14	5.03	0.981	100%
Plan 3 (two papers)	2.00	3.14	3.14	0.956	97%
Plan 4 (long span)	17.17	40.28	20.90	0.966	97%
Plan 5 (large gap)	5.86	22.10	15.83	0.935	100%

Note: High Conf. = proportion exceeding 0.80 threshold.

C.1 Verified Cases by Plan

Table C.3: Plan 0: Verified DIFFERENT cases (N=200, 6 flagged)

Author ID	Status	Verification Evidence
2806450108	Verified TRUE	Patents (household) vs agriculture; distinct domains
2830220426	Verified TRUE	Medicine: gynecology (NY, 1890s) vs toxicology (Chicago, 1920s)
2818874596	Verified TRUE	Medicine: bacteriology (Chicago) vs cardiology (Boston)
2641475794	Verified TRUE	Botany (California) vs surveying (British Columbia)
2464243470	Verified TRUE	Medical education (1890s) vs clinical medicine (1920s-1930s)

Note: Precision = 83.3% (5/6). Overall accuracy = 97.0% (194/200).

Table C.4: Plan 1: Verified DIFFERENT cases (N=200, 2 flagged)

Author ID	Status	Verification Evidence
2120931347	Verified TRUE	Pure mathematics (Europe, 1900s) vs applied engineering (USA, 1930s)

Note: Precision = 50% (1/2). Overall accuracy = 99.0% (198/200).

Table C.5: Plan 3: Verified DIFFERENT cases (N=100, 2 flagged)

Author ID	Status	Verification Evidence
369882560	Verified TRUE	Chemistry (organic synthesis, 1910) vs physics (spectroscopy, 1925)

Note: Precision = 50% (1/2).

Table C.6: Plan 4: Verified DIFFERENT cases (N=100, 5 flagged)

Author ID	Status	Verification Evidence
1966644086	Verified TRUE	Astronomy (1880s-1900s) vs geology (1920s-1930s); 40-year span
2288755076	Verified TRUE	Civil engineering (Europe) vs electrical engineering (USA); 35-year span
2525640673	Verified TRUE	Academic chemistry vs industrial chemistry patents; institutional shift
2669606853	Verified TRUE	Biology: taxonomy (1890s) vs genetics (1930s); paradigm shift

Note: Precision = 80% (4/5). Highest precision among stress tests.

D — Sample Distributions

Table D.1: Document type distribution (1880-1945)

Document Type	Count	Percentage
Journal Article	1,901,549	37.9%
Patent	1,699,511	33.9%
Unknown/Missing	1,286,354	25.7%
Book	76,234	1.5%
Conference Paper	31,456	0.6%
Book Chapter	17,715	0.4%
Total	**5,012,819**	**100.0%**

Table D.2: Patent vs non-patent author characteristics

Metric	**Patent Authors**		**Non-Patent Authors**	
	Mean	Median	Mean	Median
Papers per author	4.85	3.00	4.23	3.00
Year span (years)	1.38	0.00	1.06	0.00
Largest gap (years)	1.00	0.00	0.65	0.00
Author count	347,489		246,373	

Note: Patent authors exhibit significantly longer year spans ($t = 58.480$, $p < 0.001$) and larger gaps ($t = 95.208$, $p < 0.001$).

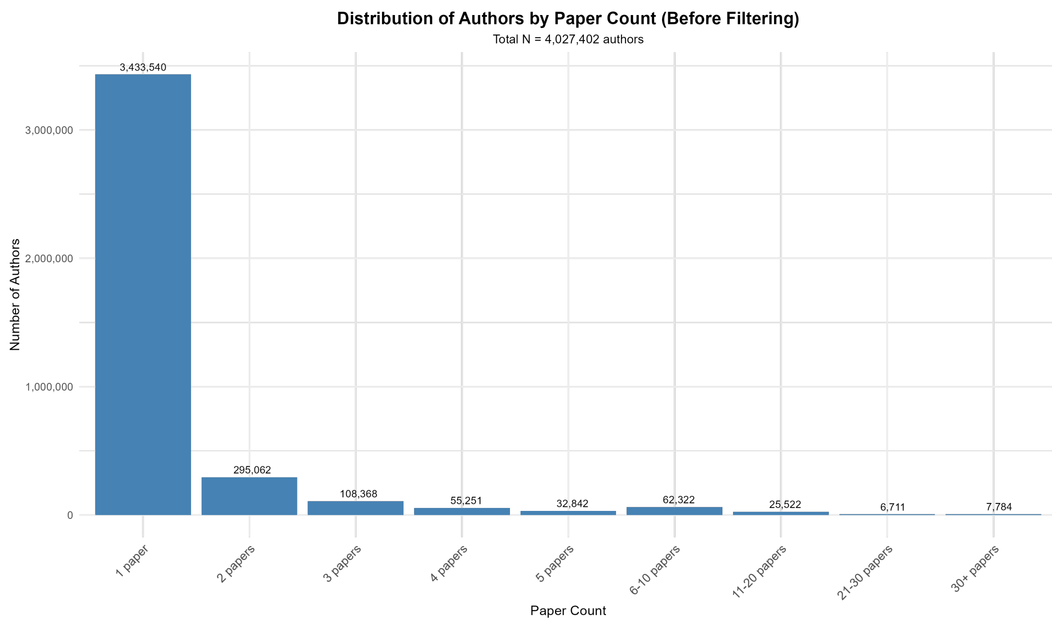

Figure D.1: Paper count distribution before filtering (N=4.0M author IDs). The overwhelming majority (85.3%) have exactly one paper, which are filtered out in Stage 2 as they require no disambiguation.

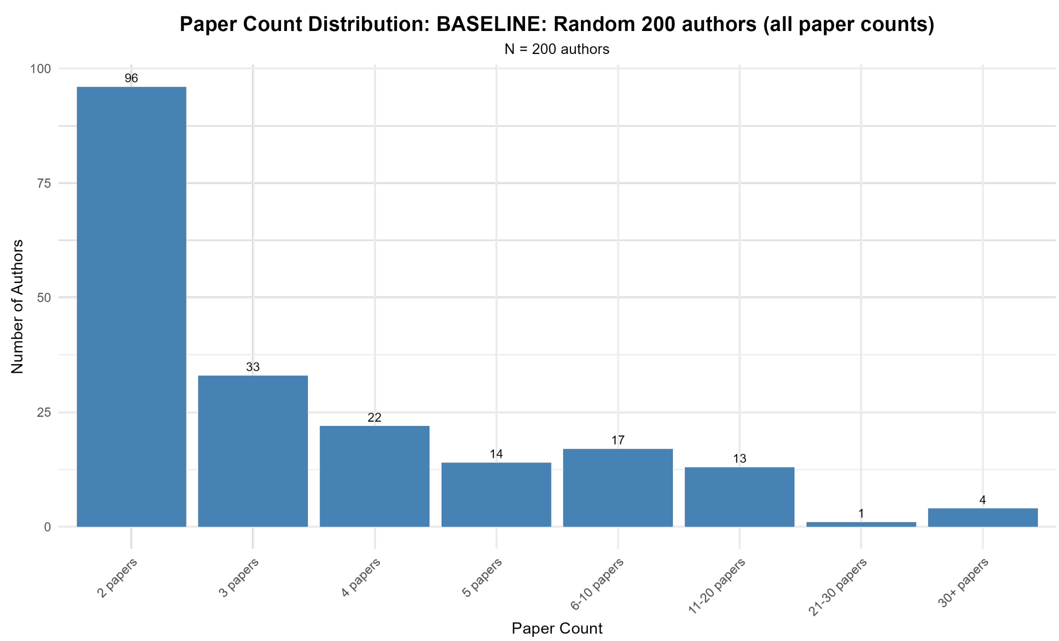

Figure D.2: Paper count distribution in evaluation sample (Plan 0, N=200). The distribution shows right-skewed pattern with most authors having 2-3 papers, and a long tail extending to highly productive authors with 30+ papers.

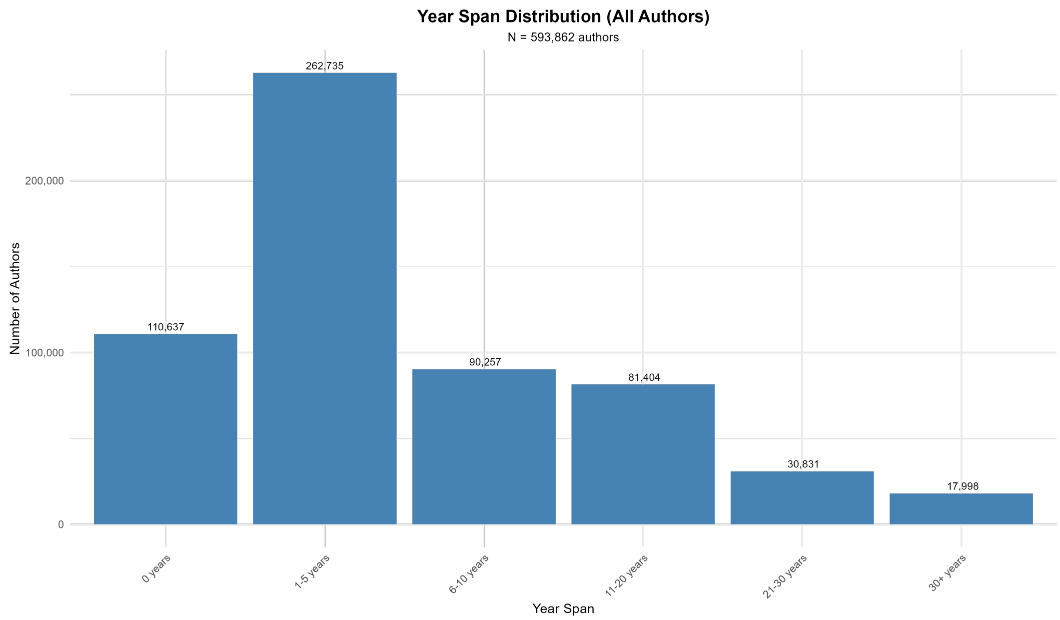

Figure D.3: Year span distribution for full sample (N=593K author IDs). The majority (88.0%) have zero span, meaning all papers were published in the same year. Non-zero spans show exponential decay, with very few authors spanning more than 30 years.

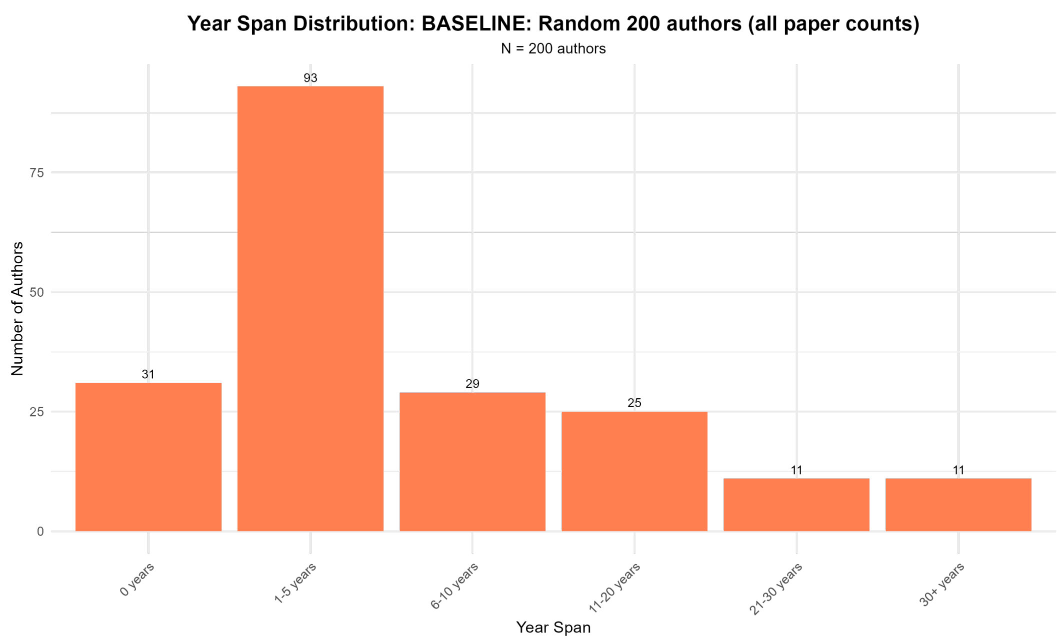

Figure D.4: Year span distribution in evaluation sample (Plan 0, N=200). Shows similar pattern to full sample but with more variation due to smaller sample size.

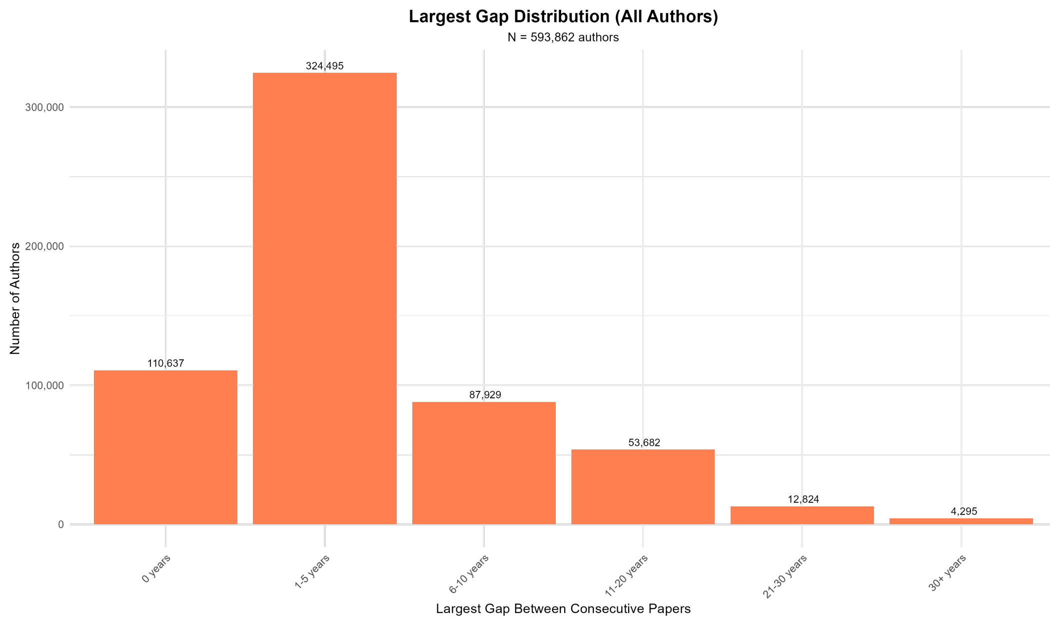

Figure D.5: Largest gap distribution for full sample (N=593K author IDs). Shows similar concentration at zero as year span, with exponential decay for positive gaps. Large gaps (¿10 years) are relatively rare, occurring in less than 3% of cases.

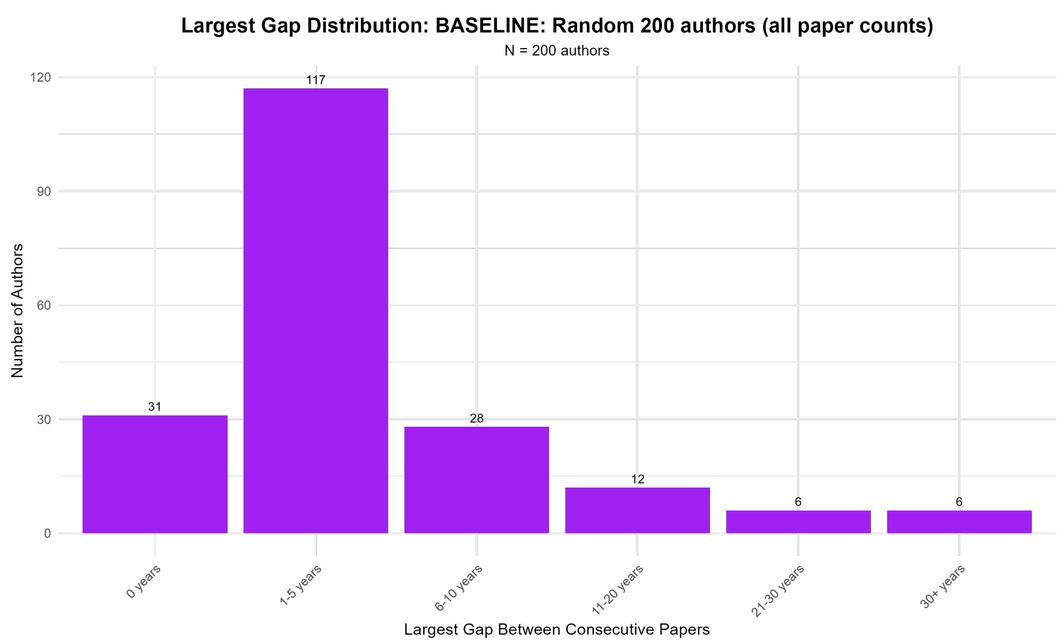

Figure D.6: Largest gap distribution in evaluation sample (Plan 0, N=200). The majority of authors show gaps of 1-5 years, with relatively few showing extreme gaps that might signal over-aggregation.

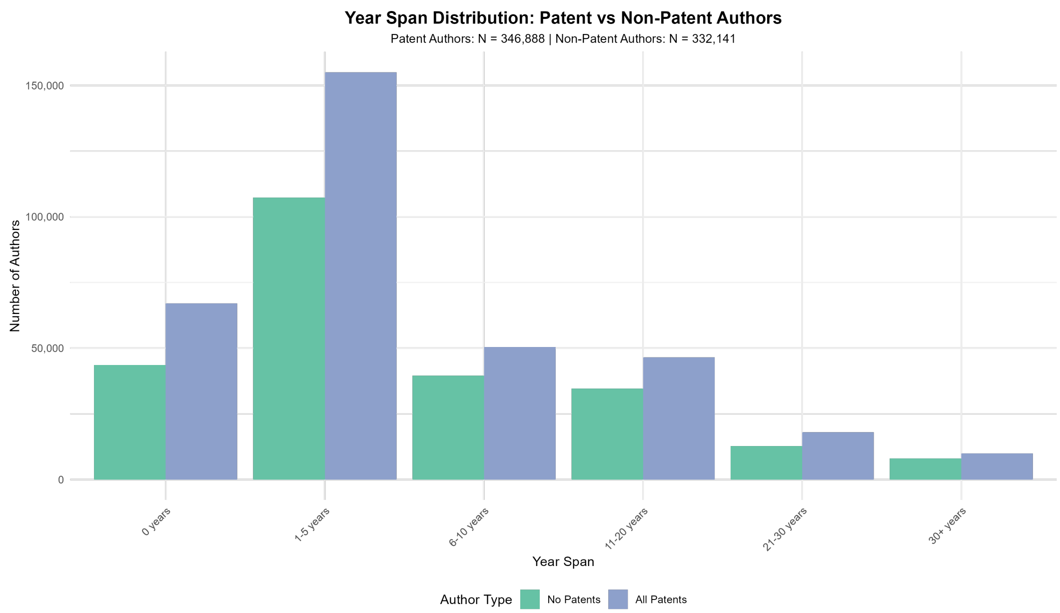

Figure D.7: Year span comparison: Patent vs Non-patent authors. Patent authors (blue) show consistently longer year spans across all ranges compared to non-patent authors (green), reflecting different publication/innovation cycles.

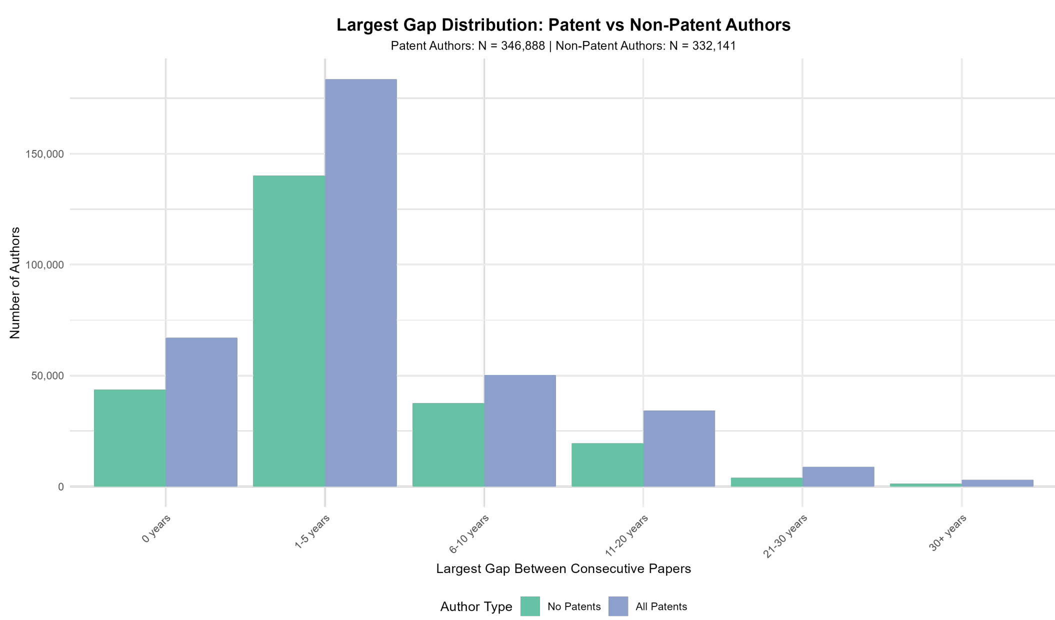

Figure D.8: Largest gap comparison: Patent vs Non-patent authors. Patent authors exhibit larger gaps between consecutive publications, with more extreme outliers in the 21-30 year and 30+ year ranges.

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